

Test of Mathematics for University Admission
Mock Set H Paper 2
Mathematical Reasoning

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

[illegible]

Question 1

For an arithmetic progression $u_1, u_2, u_3, \dots$, the condition “ u_1 and u_3 are both integers” is _____ for “every term of the progression is an integer”.

- A necessary but not sufficient
- B sufficient but not necessary
- C necessary and sufficient
- D neither necessary nor sufficient

Question 2

Consider the statement: *for every real number x , if $x^2 > x$ then $x > 1$* . How many of the following values of x are counterexamples to it?

$$-1, \quad -\frac{1}{2}, \quad 0, \quad \frac{1}{2}, \quad 1, \quad 2$$

- A 1
- B 2
- C 3
- D 4
- E 5
- F 6

Question 3

A student claims that *if n^2 is divisible by 8, then n is divisible by 8*, and argues: “Suppose $8 \nmid n$. Write $n = 8q + r$ with $1 \leq r \leq 7$. Then $n^2 = 64q^2 + 16qr + r^2$, and r^2 is one of 1, 4, 9, 16, 25, 36, 49, none of which is divisible by 8. So $8 \nmid n^2$.” Which of the following is correct?

- A The argument is valid and the claim is true.
- B The claim is true, but the argument should use remainders on division by 4.
- C The argument is invalid because $64q^2 + 16qr$ need not be divisible by 8.
- D The claim is false, and the argument fails because 16 is in the list and is divisible by 8.
- E The claim is false only when n is odd.
- F The argument proves the converse of the claim.

Question 4

Find the complete set of real values of x for which

$$x < \frac{1}{x} < x^2.$$

- A $0 < x < 1$
- B $x > 1$
- C $-1 < x < 0$
- D $x < -1$ or $x > 1$
- E $-1 < x < 0$ or $x > 1$
- F $x < -1$

Question 5

Let $f(x) = x|x| + ax$, where a is a real constant. The condition “ $a > 0$ ” is _____ for “ f is one-to-one”.

- A necessary but not sufficient
- B sufficient but not necessary
- C necessary and sufficient
- D neither necessary nor sufficient

Question 6

Four friends each make a statement about a real number x :

Alice: “ $x > 2$.” Bob: “ $x > 3$.” Carol: “ $x < 5$.” Dan: “ $x^2 > 9$.”

Exactly two of them are telling the truth. Find the complete set of possible values of x .

- A $2 < x \leq 3$
- B $x < -3$
- C $x < -3$ or $2 < x \leq 3$
- D $x < -3$ or $2 < x < 5$
- E $-3 \leq x \leq 3$
- F $x \geq 5$

Question 7

On an island every inhabitant either always tells the truth or always lies. A group of n inhabitants, where $n \geq 3$, sit around a round table, and each of them says: “The person on my left is a liar.” Which of the following are true?

- I When $n = 6$ this is possible.
 - II When $n = 7$ this is impossible.
 - III Whenever this is possible, exactly half of the people are liars.
 - IV Whenever this is possible, there is only one way to decide who is a liar.
- A I, II and III only
 - B I and II only
 - C I, II, III and IV
 - D II and III only
 - E I only
 - F I, II and IV only

Question 8

For real numbers a, b, c , consider the statement G : “ a, b, c (in that order) are consecutive terms of a geometric progression with non-zero common ratio”. Which of the following are true?

- I $b^2 = ac$ is necessary for G .
- II $b^2 = ac$ is sufficient for G .
- III $b^2 = ac$ together with $a \neq 0$ is sufficient for G .
- IV $b^2 = ac$ together with $b \neq 0$ is sufficient for G .

- A I only
- B I and II only
- C I and III only
- D I, III and IV only
- E I and IV only
- F I, II, III and IV

Question 9

How many of the following statements about integers are true?

- I For every integer a there is an integer b such that $a + b$ is odd and ab is even.
- II There is an integer b such that $a + b$ is odd for every integer a .
- III For every integer a there is an integer b such that $a^2 + b^2 = 2ab + 1$.
- IV There is an integer a such that ab is even for every integer b .
- V For all integers a and b , if $a + b$ is odd then ab is even.

- A 1
- B 2
- C 3
- D 4
- E 5

Question 10

A counter starts at 0 on a number line. Each move either adds 3 to its position or subtracts 5 from it. Which of the following are true?

- I Every integer can be reached.
- II The least number of moves needed to reach 1 is 3.
- III The integer 2 can be reached in fewer than 4 moves.
- IV After exactly 10 moves the counter cannot be at 0.

- A I only
- B I and II only
- C II and IV only
- D I, II, III and IV
- E I and IV only
- F I, II and IV only

Question 11

Let S be the set of points (x, y) in the plane satisfying

$$|x| + |y| = |x + y| + 1.$$

Which of the following are true?

- I** S is unbounded.
 - II** S contains the point $(\frac{1}{2}, -\frac{1}{2})$.
 - III** Every point of S satisfies $xy < 0$.
 - IV** S contains a point with $|x| + |y| < 1$.
- A** I and II only
 - B** I, II and III only
 - C** II and III only
 - D** I, II, III and IV
 - E** I and III only
 - F** III only

Question 12

Three arguments are offered for the claim that $n^5 - n$ is divisible by 5 for every positive integer n .

- I.** For $n = 1$ it is 0. Expanding, $((n + 1)^5 - (n + 1)) - (n^5 - n) = 5n^4 + 10n^3 + 10n^2 + 5n$, a multiple of 5. So each value differs from the previous one by a multiple of 5, and every value is a multiple of 5.
- II.** $n^5 - n$ modulo 5 depends only on n modulo 5, and for $n \equiv 0, 1, 2, 3, 4$ it is 0, 0, 30, 240, 1020 modulo 5, all $\equiv 0$.
- III.** $n^5 - n = (n - 1)n(n + 1)(n^2 + 1)$, which is a product of five consecutive integers and so is divisible by 5.

Which of these are valid proofs?

- A** I and II only
- B** I only
- C** II only
- D** I, II and III
- E** II and III only
- F** I and III only

Question 13

Which of the following statements are true?

- I** There is a polynomial p of degree 3 with $p(0) = 0$, $p(1) = 1$, $p(2) = 4$ and $p(3) = 10$.
 - II** There is a polynomial p of degree at most 2 with $p(0) = 0$, $p(1) = 1$, $p(2) = 4$ and $p(3) = 10$.
 - III** There are infinitely many polynomials p of degree 4 with $p(0) = 0$, $p(1) = 1$, $p(2) = 4$ and $p(3) = 10$.
 - IV** There is a polynomial p with integer coefficients and $p(0) = 0$, $p(1) = 1$, $p(2) = 4$ and $p(3) = 10$.
- A** I only
 - B** I, III and IV only
 - C** I and III only
 - D** I, II, III and IV
 - E** III only
 - F** I and IV only

Question 14

Consider the four sets of real numbers

$$A = \{x : |x| < \frac{1}{n} \text{ for every positive integer } n\}, \quad B = \{x : |x| < \frac{1}{n} \text{ for some positive integer } n\},$$

$$C = \{x : x < n \text{ for every positive integer } n\}, \quad D = \{x : x > n \text{ for some positive integer } n\}.$$

Which of the following are true?

- I** $A = \{0\}$
 - II** B is the interval $-1 < x < 1$.
 - III** C is the set of all real numbers.
 - IV** D is the set of all real numbers.
- A** I only
 - B** I, II, III and IV
 - C** II and III only
 - D** I and III only
 - E** I and II only
 - F** I, II and IV only

Question 15

The function f is continuous on $[0, 1]$, satisfies $0 \leq f(x) \leq 1$ for every x , and $\int_0^1 f(x) dx = \frac{1}{2}$. Which of the following must be true?

- I $\int_0^1 (f(x))^2 dx \leq \frac{1}{2}$
- II $\int_0^1 (f(x))^2 dx \geq \frac{1}{4}$
- III $\int_0^1 xf(x) dx \leq \frac{3}{8}$
- IV $\int_0^1 xf(x) dx = \frac{1}{4}$
- A I only
- B I and II only
- C III only
- D I, II and III only
- E I, II, III and IV
- F II and III only

Question 16

For each positive integer n , let $d(n)$ be the highest common factor of $n^2 + 1$ and $(n + 1)^2 + 1$. Which of the following are true?

- I $d(n)$ is always either 1 or 5.
- II $d(2) = 5$.
- III $d(n) = 5$ for infinitely many n .
- IV $d(n) = 2$ for some n .
- A I only
- B I, II and III only
- C I and II only
- D I, II, III and IV
- E II and III only
- F I and III only

Question 17

A continuous function f satisfies $f(x + 1) = f(x) + 1$ for every real x . Which of the following must be true?

- I $f(n) = n + f(0)$ for every integer n .
- II The function $g(x) = f(x) - x$ is bounded.
- III f is increasing.
- IV If $f(0) = 0$, then $f(x) \geq x - 1$ for every x .
- A I and II only
- B I only
- C I, II and III only
- D I, II and IV only
- E I, II, III and IV
- F II only

Question 18

Which of the following statements are true for every real number x ? (Angles are in radians.)

- I $|\sin x| + |\cos x| \geq 1$
- II $\tan x + \frac{1}{\tan x} \geq 2$, whenever both terms are defined.
- III $|\sin x - \cos x| \leq \sqrt{2}$
- IV $\cos(\sin x) > \sin(\cos x)$
- A I and III only
- B I, II and III only
- C I, III and IV only
- D I, II, III and IV
- E III and IV only
- F I and IV only

Question 19

Positive real numbers x , y and z satisfy $xyz = 1$. Which of the following must be true?

- I $x + y + z \geq 3$
- II $xy + yz + zx \geq 3$
- III $x^2 + y^2 + z^2 \geq x + y + z$
- IV $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq x + y + z$
- A I only
- B I and II only
- C I, II, III and IV
- D II and III only
- E I and IV only
- F I, II and III only

Question 20

A quartic polynomial f has positive leading coefficient and satisfies

$$f(0) = 0, \quad f'(0) = 0, \quad f(1) = -1, \quad f(2) = 0.$$

Which of the following must be true?

- I f has exactly one local minimum.
- II $f(x) < 0$ for every x with $0 < x < 2$.
- III f has at least two stationary points.
- IV $f(-1) > 0$
- A I and III only
- B II and III only
- C III only
- D I, II and III only
- E I, II, III and IV
- F III and IV only

END OF PAPER